

PRODUCTION AI ENGINEERING · T5 · REAL VS SIMULATED REAL VS SIMULATED

T5 Real vs Simulated: Observability & Governance POC

What in the POC is real, what is simulated, and what is recorded, so every number can be weighed for what it is.

T5

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2 min read · Published 2026-09-30 · Real vs simulated

Real

Component	What is real about it
Processes	Every agent runtime, the deployment API and the approvers run as separate OS processes (62 in the run).
Crashes	<code>SIGKILL</code> sent by the process to itself at a named point (2 in the run); no cleanup runs; the harness restarts it.
Network	HTTP over localhost sockets; the lost response is a real client socket timeout after 1,503 ms.
Durability	SQLite with WAL for workflow checkpoints, approvals, evidence and the deployment API's records; the evidence store uses <code>synchronous=FULL</code> .
Idempotency	The deployment API stores responses by <code>Idempotency-Key</code> and replays them; 409 and 422 behaviour is implemented and tested.
Retries	The gateway retries on timeout, 503 and 409 with the same key; attempts are numbered durably.
OpenTelemetry	The OpenTelemetry Python SDK 1.45.0: real spans, W3C <code>traceparent</code> propagation into the deployment API, log correlation, metrics.
Evidence integrity	SHA-256 hash chain across concurrent writers; witness anchors; verification and four tampering attempts.
Policy and approvals	Deterministic versioned policy engine with document digests; HMAC-signed, action-bound approval decisions with eligibility checks.
Model	qwen3:8b served by Ollama 0.30.11, 30 calls in the scenarios plus 20 in the drift probe, recorded on tape.
Tests	31 of 31 pass, including end-to-end scenario tests with real processes.

Simulated

Component	How it is simulated	What that does not show
Deployment API	A small HTTP service with revision numbers, idempotency and fault injection	asynchronous rollouts, partial application, eventual consistency of reads
Faults	Injected from a per-scenario fault script: reject before commit, acknowledge without applying, drop the response after commit	the variety and timing of real failures
The incident	INC-4471 and INC-4472, with fixed signals	real telemetry, real diagnosis
People	A scripted operator process approves or rejects after 0.4 s	approval latency, fatigue, human error
Identities and credentials	Configured principals, a static bearer credential, SPIFFE-shaped names	real token exchange (T1 covers that)
Compromised plan (E7)	The proposal is replaced by a scale-to-zero request and a direct gateway call	how a real model gets compromised
Duplicate delivery (E6)	The runtime invokes the tool step a second time after its first completion	a real queue's redelivery timing
External witness	A separate file	WORM storage, a transparency log, separate control
Retention and access control	Declared per event type in config; not enforced by a store	enforcement, deletion, access audit

Recorded

What	Where	Replay
Every model call's request hash and full response	<code>scenarios/<scenario>/tape/<role>/model_tape.jsonl</code>	lineage run replay 2026-09-30-recorded : 60 answers, 0 misses, identical yes
The model server's model list, including the digest	the same tapes (<code>/api/tags</code>)	replayed with the calls
The drift probe's calls	<code>drift/tape/model_tape.jsonl</code>	replayed identically: yes

Everything that isn't the model is re-executed on replay: processes, HTTP, SQLite, OpenTelemetry, hashing, scoring. The replay is a rerun with the model's answers held fixed, not a playback of results.

Evidence

Method

Observability

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Production AI Engineering · T5 · Real vs simulated. Every measured number here is substituted at build time from observability_governance_poc/runs/2026-09-30-recorded/facts.json; the enterprise systems are simulated and the run is deterministic. Figures carry their provenance.

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